MIL-S-50218B (AR)
13 February 1978
SUPERSEDING
MIL-S-50218A (MU)
30 October 1975

MILITARY SPECIFICATION

SWITCH, NOSE CRUSH ASSEMBLY,

ASSEMBLING AND PACKING

-(DRAGON)

This specification is approved for use by the US Army Armament Research and Development Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification contains requirements not covered by the drawings and provides quality assurance provisions for the fabrication of parts, assembly and packing of one type of nose crush switch.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents.-The following documents of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-C-5541 - Chemical Conversion Coating on Aluminum and Aluminum Alloys
MIL-E-45782 - Electrical Wiring, Procedure for
MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification for

STANDARDS

MILITARY

MIL-STD-1169 - Packaging, Packing and Marking for Shipment of Inert Ammunition Components.
MIL-STD-1460 - Soldering of Electrical Connections and Printed Wiring Assemblies Procedure for

FSC: 1336

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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FEDERAL TEST METHOD STANDARDS

No. 141 - Paint, Varnish, Lacquer, and Related Materials; Method of Inspection, Sampling, and Testing.

DRAWINGS (see 6.4)

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND

9220940 - Nose Crush Switch Assembly

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 Materials.-Materials shall be in accordance with applicable drawings and specifications.

3.2 Components and assemblies.-The components and assemblies shall comply with all requirements specified on drawing 9220940 and associated drawings and with all requirements of applicable standards and specifications.

3.3 Electrical continuity and insulation resistance of Nose Crush Switch Assembly (Dwg. 9220940) and Components.

3.3.1 Electrical continuity of each wire.

3.3.1.1 Electrical resistance of black ground wire from one terminal in the ogive (dwg. 9220941) to the other end of the wire shall not exceed 0.2 ohm. This requirement shall also apply to measurements taken on intermediate assemblies between the terminal lug at one end and the other end of the wire.

3.3.1.2 Electrical resistance of the red live wire from one terminal in the inner cone (dwg. 9220942) to the other end of the wire shall not exceed 0.2 ohm. This requirement shall also apply to measurements taken on intermediate assemblies between the terminal lug at one end and the other end of the wire.



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3.3.2 Insulation (open circuit) resistance of the nose crush switch and wires (dwg. 9220940).-The wire-to-wire electrical resistance between ends of the black and red wire, or between the inner cone and the ogive shall not be less than 500 megohms.

3.3.3 Tensile strength of terminal assembly.-The terminal assembly (see Dwg. 9317968) shall meet the requirements of MIL-C-45732 para. 3.4.2.1, when subjected to a tensile strength test as specified in 4.5.4 herein.

3.4 Protective finishes.-The assemblies and test panels specified below shall be subjected to and must withstand the protective finish tests outlined below without failure. Test panels must be of the same metal, same surface finish and coated identically to and concurrently with the manufactured parts they represent.

3.4.1 Corrosion resistance.-Chemical film-test panels, representing both, Missile Ogive, Dwg. 9220941 and Inner Cone, Dwg. 9220942 shall be subjected to a 100 hour 5 percent salt spray test in accordance with MIL-C-5541 and Method 6061 of Fed Test Method STD. No. 141. The test panels shall meet the corrosion resistance requirements specified in MIL-C-5541.

3.4.2 Paint adhesion.-Epoxy painted test panels representing Nose Crush Switch Assembly, Dwg. 9220940 and epoxy-painted items of Nose Crush Switch Assembly, Dwg. 9220940 shall be subjected to paint adhesion knife, and tape tests in accordance with MIL-C-5541 and Methods 6301.1 and 6304.1 of Federal Test Method STD. 141. The paint on the test panels and items shall not exhibit flaking, chipping, peeling, loss of adhesion or discoloration.

3.5 First article inspection.-This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.6 Workmanship.-All parts and assemblies shall be fabricated, assembled and finished in a thorough, workmanlike manner as defined in 3.6.1, 3.6.2, and 3.6.3.

3.6.1 Fabrication and assembly.-The parts and assemblies shall be complete and shall be free of burrs, sharp or chipped edges, cracks, dents, gouges, toolmarks, holes, inclusions, blowholes, blisters, warpage or other distortion, porosity, rust, corrosion or any other undesirable characteristics, signs of damage or poor workmanship.



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3.6.2 Finishes.-Surface coatings and finishes of parts and assemblies shall be continuous and complete and shall not have scratches, blisters, porosity, corrosion or other signs of defective finish, or damage thereto, except for superficial scratches that do not expose the base metal.

3.6.3 Cleanliness.-The parts shall be clean and free from dirt, corrosion, chips, grease, contaminants or other foreign matter. The cleaning method and agent used shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agent.

## 4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions.-Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of inspections.-The following types of inspection shall be conducted on this item:

a. First article inspection

b. Quality conformance inspection

4.3 First article inspection

4.3.1 Submission.-The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of the following items in sample quantities as indicated.



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| Drawings | Part Description | No. of Units | Test Panels |
|---|---|---|---|
| 9220942 | Inner Cone | 6 | |
| | Inner Cone - Protective Finish: Chromate Film Test Panels | | 15 |
| 9220941 | Ogive, Missile | 6 | |
| 9220944 | Post, Insulator | 6 | |
| 9220940 | Nose Crush Switch Assembly | 8 | |
| | Ogive external protective finish | | 15 |
| 9216784 | Wireway | 6 | |
| 9317967-1 | Wire, Electrical | 6 | |
| 9317967-2 | Wire, Electrical | 6 | |
| 9317968-1 | Terminal assembly | 6 | |
| 9317968-2 | Terminal assembly | 6 | |
| 9317969 | Ogive, Missile Subassembly | 6 | |
| 9317970 | Cone, Inner Subassembly | 6 | |
| 9317983 | Grommet, Rubber | 6 | |
| | Complete sets of other parts not listed above, such as washers, rivets, etc. | 6 of each | |

4.3.2 Inspections to be performed.-See MIL-A-48078 and Table I specified herein. Sequence of testing shall be in accordance with Figures 1 and 2.

4.3.3 Rejection.-See MIL-A-48078.



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TABLE I - FIRST ARTICLE INSPECTION

## CLASSIFICATION OF DEFECTS & TESTS

MIL-S-50218B (AR)

| PARAGRAPH | TITLE | | SHEET 1 OF 2 | | DRAWING NUMBER |
|---|---|---|---|---|---|
| | Nose Crush Switch Assembly and Components | | | | See Below |
| | | | | | NEXT HIGHER ASSEMBLY |
| CATEGORY | EXAMINATION OR TEST | NO. OF SAMPLE UNITS | AQL OR 100% | REQUIREMENT PARAGRAPH | PARAGRAPH REFERENCE /INSPECTION METHOD |
| | Inner Cone (dwg 9220942) | | | | |
| | (See Figure 1A for sequence of testing) | | | | |
| | Examination for Defects | 6 | | 3.2 | 4.4.2.1 |
| | Salt Spray Test on: | | | | |
| | Test Panels | 13 | | 3.4.1 | 4.5.3.1 |
| | Ogive, Missile (dwg 9220941) | | | | |
| | (See Figure 1B for sequence of testing): | | | | |
| | Examination for Defects | 6 | | 3.2 | 4.4.2.2 |
| | Terminal Assembly (dwg 9317968-1 & 9317968-2) | | | | |
| | (See Figure 2 for sequence of testing) | | | | |
| | Examination for Defects | 6each | | 3.2 | 4.4.2.3 |
| | Tensile Strength on Terminal Lugs | | | | |
| | (Visual examination for damage) | 6each | | 3.3.3 | 4.5.4. |
| | Electrical continuity between terminal lug and end of black wire | 6each | | 3.3.1.1 | 4.5.2.1.a |
| | Electrical continuity between terminal lug and end of red wire | 6each | | 3.3.1.2 | 4.5.2.1.a |
| | Ogive, Missile Subassembly (dwg 9317969) | | | | |
| | Examination for Defects | 6 | | 3.2 | 4.4.2.4 |
| | Cone, Inner Subassembly (dwg 9317970) | | | | |
| | Examination for Defects | 6 | | 3.2 | 4.4.2.5 |

NOTES:
All defects listed in Table I are classified as Major unless otherwise specified.



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TABLE I - FIRST ARTICLE INSPECTION

## CLASSIFICATION OF DEFECTS & TESTS

MIL-S-50218B (AR)

| PARAGRAPH | TITLE: Nose Crush Switch Assembly and Components | | SHEET 2 OF 2 | | DRAWING NUMBER / NEXT HIGHER ASSEMBLY |
|---|---|---|---|---|---|
| CATEGORY | EXAMINATION OR TEST | NO. OF SAMPLE UNITS | AQL OR 100% | REQUIREMENT PARAGRAPH | PARAGRAPH REFERENCE /INSPECTION METHOD |
| | Post Insulator (dwg 9220944) | | | | |
| | Examination for Defects | 6 | | 3.2 | 4.4.2.6 |
| | Nose Crush Switch Assembly (dwg 9220940) | | | | |
| | Examination of Defects | 8 | | 3.2 | 4.4.2.7 |
| | Paint Adhesion - Knife test on: | | | | |
| | N.C.S. Assemblies | 8 | | 3.4.2 | 4.5.3.2.1 |
| | Test Panels | 11 | | 3.4.2 | 4.5.3.2.1 |
| | Paint - Adhesion - Tape Test on: | | | | |
| | N.C.S. Assemblies | 8 | | 3.4.2 | 4.5.3.2.1 |
| | Test Panels | 11 | | 3.4.2 | 4.5.3.2.1 |
| | Electrical Continuity between: | | | | |
| | a. Ogive to end of black wire | 8 | | 3.3.1.1 | 4.5.2.1.b |
| | b. Inner cone to end of red wire | 8 | | 3.3.1.2 | 4.5.2.1.b |
| | Insulation Resistance | 8 | | 3.3.2 | 4.5.2.2 |
| | Wireway (dwg 9216784) | | | | |
| | Examination for Defects | 6 | | 3.2 | 4.4.2.8 |
| | Other parts | | | | |
| | Examination for Defects | 6 each | | 3.2 | 4.4.2.9 |

NOTES:

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FIGURE 1

FIRST ARTICLE TEST

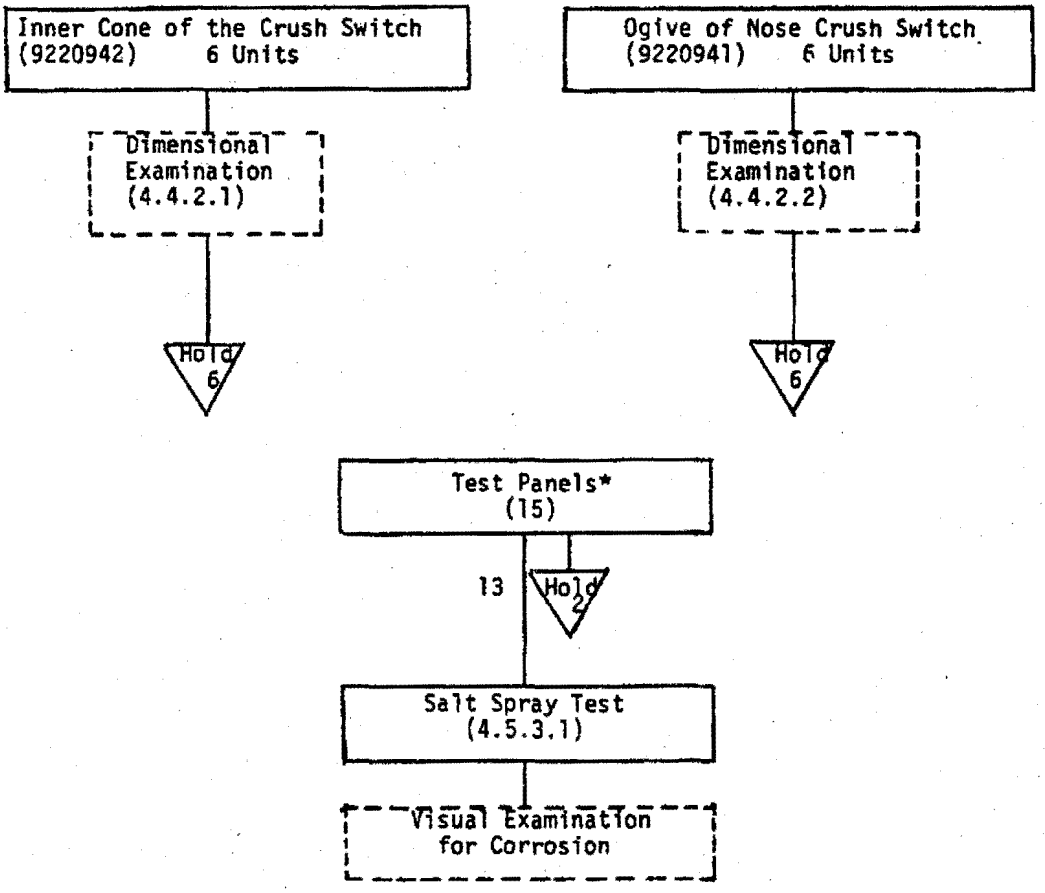



Figure 1A

Figure 1B

*NOTE: Test panels represent both Inner Cone and Ogive



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FIRST ARTICLE TEST

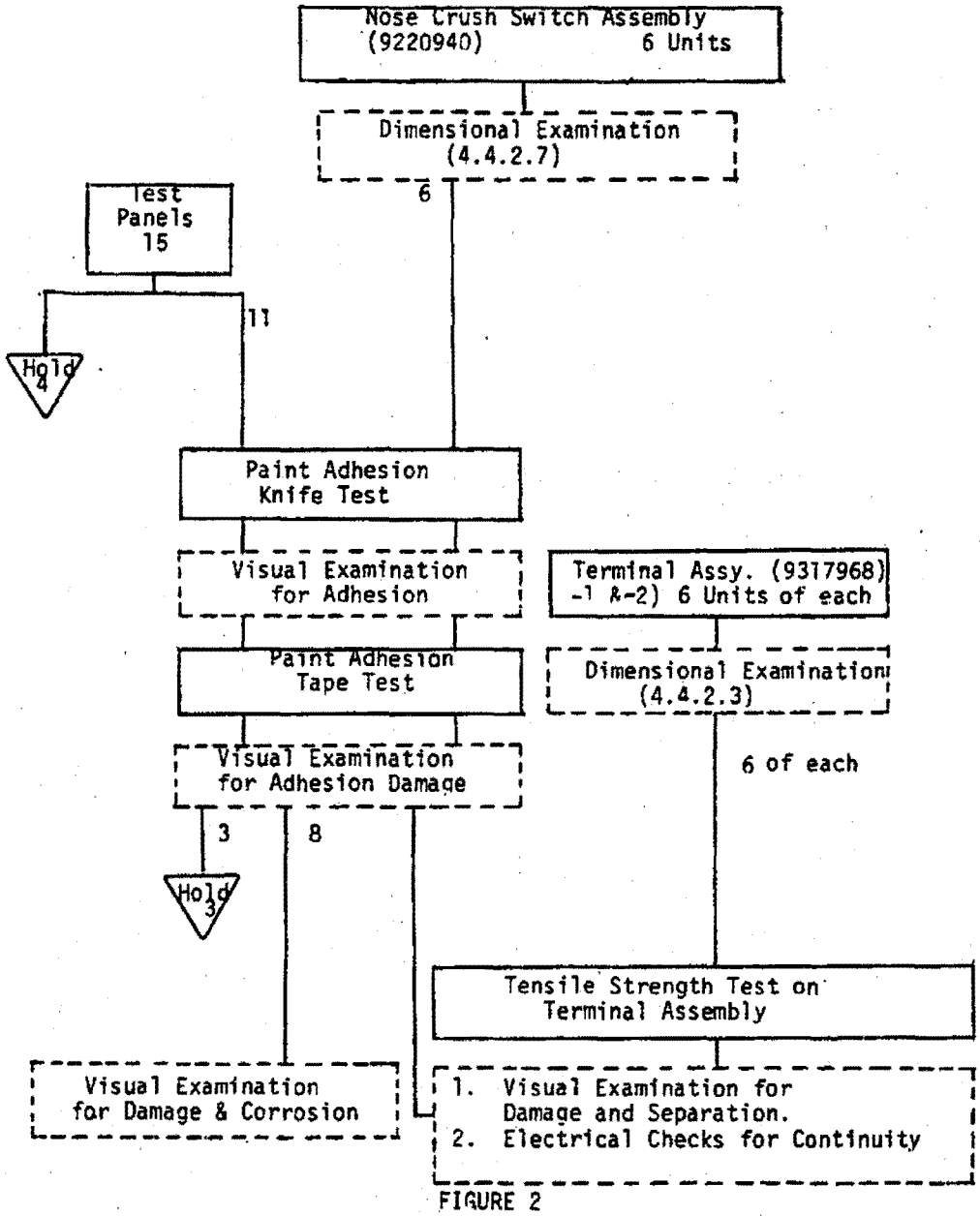



FIGURE 2



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4.4 Quality conformance inspection

4.4.1 Inspection lot formation.-Inspection lots shall comply with the lot formation provisions of MIL-A-48078. In addition, inspection lots of Nose Crush Switch Assemblies shall contain:

a. Cone, Inner (Dwg. 9220942) from not more than one interfix number from one manufacturer.

b. Ogive, Missile (Dwg. 9220941) from not more than one interfix number from one manufacturer.

c. Terminal Assembly (Dwg. 9317968-1 & 9317968-2) from not more than one interfix number from one manufacturer.

d. Post, Insulator (Dwg. 9220944) from not more than one interfix number from one manufacturer.

e. Washer, Nonmetallic (Dwg 9233180) from not more than one interfix number from one manufacturer.

4.4.2 Examination.-See MIL-A-48078.

a. Sampling plans.-Unless otherwise specified in the Classification of Defects and Test tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.



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## CLASSIFICATION OF DEFECTS & TESTS

MIL-S-502183 (AS)

| PARAGRAPH | TITLE | | SHEET 1 OF 1 | | DRAWING NUMBER 9220942 |
|---|---|---|---|---|---|
| 4.4.2.1 | Cone, Inner | | | | NEXT HIGHER ASSEMBLY 9317975 |
| CATEGORY | EXAMINATION OR TEST | NO. OF SAMPLE UNITS | AQL OR 100% | REQUIREMENT PARAGRAPH | PARAGRAPH REFERENCE /INSPECTION METHOD |
| Critical | None defined | | | | |
| Major | | | | | |
| 101 | Wall Thickness | | 0.40% | 3.2 | Gage |
| 102 | Outside diameter, mounting end | | 0.40% | 3.2 | Gage |
| 103 | Overall height | | 0.40% | 3.2 | Gage |
| 104 | Contour or configuration of nose | | 0.40% | 3.2 | Gage |
| 105 | Location of mounting holes to ogive | | 0.40% | 3.2 | Gage |
| 106 | Diameter of mounting holes | | 0.40% | 3.2 | Gage |
| Minor | | | | | |
| 201 | Diameter of rivet hole | | 0.65% | 3.2 | Gage |
| 202 | Diameter of grommet holes | | 0.65% | 3.2 | Gage |
| 203 | Protective finish inadequate, scratched or damaged | | 0.65% | 3.6.2 | Visual |
| 204 | Evidence of poor workmanship | | 1.00% | 3.6 | Visual |

NOTES:

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## CLASSIFICATION OF DEFECTS & TESTS

| PARAGRAPH | TITLE | | SHEET 1 OF 2 | | DRAWING NUMBER |
|---|---|---|---|---|---|
| 4.4.2.2 | Ogive, Missile | | | | 9220941 |
| | | | | | NEXT HIGHER ASSEMBLY 9317969 |
| CATEGORY | EXAMINATION OR TEST | NO. OF SAMPLE UNITS | AQL OR 100% | REQUIREMENT PARAGRAPH | PARAGRAPH REFERENCE /INSPECTION METHOD |
| Critical | None defined | | | | |
| Major | | | | | |
| 101 | Inside diameter, Warhead body mating surface | | 0.40% | 3.2 | Gage |
| 102 | Overall height | | 0.40% | 3.2 | Gage |
| 103 | Location of warhead body mounting holes | | 0.40% | 3.2 | Gage |
| 104 | Diameter of warhead body mounting holes | | 0.40% | 3.2 | Gage |
| 105 | Wall thickness - sides | | 0.40% | 3.2 | Gage |
| 106 | Nose apex configuration and contour and wall thickness at apex | | 0.40% | 3.2 | Gage |
| 107 | Roundness of warhead mounting end | | 0.40% | 3.2 | Gage |
| 108 | Diameter of inner cone mounting holes | | 0.40% | 3.2 | Gage |
| 109 | Location of inner cone mounting holes | | 0.40% | 3.2 | Gage |
| 110 | Diameter of hole for grounding connection | | 0.40% | 3.2 | Gage |
| 111 | Location of hole for grounding connection | | 0.40% | 3.2 | Gage |
| Minor | | | | | |
| 201 | Perpendicularity of mounting end to sides | | 0.65% | 3.2 | Gage |
| 202 | Location of wireway slot | | 0.65% | 3.2 | Gage |
| 203 | Dimensions of wireway slot | | 0.65% | 3.2 | Gage |
| 204 | Protective finish inadequate, scratched or damaged | | 0.65% | 3.6.2 | Visual |
| 205 | Evidence of poor workmanship | | 1.00% | 3.6 | Visual |

NOTES:





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## CLASSIFICATION OF DEFECTS & TESTS

MIL-S-502183 (AR)

| PARAGRAPH | TITLE | | SHEET 1 OF 1 | | DRAWING NUMBER 9317968-1 9317968-2 |
|---|---|---|---|---|---|
| 4.4.2.3 | Terminal Assembly | | | | NEXT HIGHER ASSEMBLY 9317970 9317969 |
| CATEGORY | EXAMINATION OR TEST | NO. OF SAMPLE UNITS | AQL OR 100% | REQUIREMENT PARAGRAPH | PARAGRAPH REFERENCE /INSPECTION METHOD |
| Critical | None defined | | | | |
| Major | | | | | |
| 101 | Tensile strength on terminal assembly (2) | (a) each | 4.00% | 3.3.3 | 4.5.4 |
| 102 | Electrical continuity between terminal lug and end of black wire | | 100% | 3.3.1.1 | 4.5.2.1.a |
| 103 | Electrical continuity between terminal lug and end of red wire | | 100% | 3.3.1.2 | 4.5.2.1.a |
| 104 | Insulation on either wire damaged or defective | | 0.40% | 3.2 | Visual |
| Minor | | | | | |
| 201 | Length of wire | | 0.65% | 3.2 | Gage |
| 202 | End of wire not properly tinned | | 0.65% | 3.2 | Visual |
| 203 | Evidence of poor workmanship | | 1.00% | 3.6 | Visual |

NOTES:
(a) Sample size per MIL-STD-105, Inspection Level 1.

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## CLASSIFICATION OF DEFECTS & TESTS

MIL-STD-50123B (AR)

| PARAGRAPH | TITLE | | SHEET 1 OF 1 | | DRAWING NUMBER |
|---|---|---|---|---|---|
| 4.4.2.4 | Ogive, Missile Subassembly | | | | 2317239 |
| | | | | | NEXT HIGHER ASSEMBLY |
| | | | | | 9220340 |
| CATEGORY | EXAMINATION OR TEST | NO. OF SAMPLE UNITS | AQL OR 100% | REQUIREMENT PARAGRAPH | PARAGRAPH REFERENCE /INSPECTION METHOD |
| Critical | None defined | | | | |
| Major | | | | | |
| 101 | Location of terminal assembly | | 0.40% | 3.2 | Gage |
| 102 | Terminal assembly incomplete - any part missing or defective | | 100% | 3.2 | Visual |
| 103 | Flat washer not in 360° contact with metal portion of terminal lug | | 100% | 3.2 | Visual |
| 104 | Rivet roll over not 360° | | 100% | 3.2 | Visual |
| Minor | | | | | |
| 201 | Protective finish inadequate, scratched or damaged | | 0.65% | 3.6.2 | Visual |
| 202 | Washer not coated with varnish after assembly | | 0.65% | 3.6.2 | Visual |
| 203 | Evidence of poor workmanship | | 1.00% | 3.6 | Visual |

NOTES:



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QUALITY CONFORMANCE INSPECTION

## CLASSIFICATION OF DEFECTS & TESTS

MIL-S-50213B (AR)

| PARAGRAPH | TITLE | | SHEET 1 OF 1 | | DRAWING NUMBER 9317970 |
|---|---|---|---|---|---|
| 4.4.2.5 | Cone, Inner Subassembly | | | | NEXT HIGHER ASSEMBLY 9220940 |
| CATEGORY | EXAMINATION OR TEST | NO. OF SAMPLE UNITS | AQL OR 100% | REQUIREMENT PARAGRAPH | PARAGRAPH REFERENCE /INSPECTION METHOD |
| Critical | None defined | | | | |
| Major | | | | | |
| 101 | Location of terminal assembly | | 0.40% | 3.2 | Visual |
| 102 | Terminal assembly incomplete - any part missing or defective | | 100% | 3.2 | Visual |
| 103 | Flat washer not in 360° contact with metal portion of terminal lug | | 100% | 3.2 | Visual |
| 104 | Rivet roll over not 360° | | 100% | 3.2 | Visual |
| Minor | | | | | |
| 201 | Protective finish inadequate, scratched or damaged | | 0.65% | 3.6.2 | Visual |
| 202 | Washer not coated with varnish after assembly | | 0.65% | 3.6.2 | Visual |
| 203 | Evidence of poor workmanship | | 1.00% | 3.6 | Visual |

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## CLASSIFICATION OF DEFECTS & TESTS

MIL-S-50218B (AR)

| PARAGRAPH | TITLE | | SHEET 1 OF 1 | | DRAWING NUMBER 9220944 |
|---|---|---|---|---|---|
| 4.4.2.6 | Post, Insulator | | | | NEXT HIGHER ASSEMBLY 9220940 |
| CATEGORY | EXAMINATION OR TEST | NO. OF SAMPLE UNITS | AQL OR 100% | REQUIREMENT PARAGRAPH | PARAGRAPH REFERENCE /INSPECTION METHOD |
| Critical | None defined. | | | | |
| Major | | | | | |
| 101 | Length of collar | | 0.40% | 3.2 | Gage |
| 102 | Minor diameter | | 0.40% | 3.2 | Gage |
| 103 | Hole diameter | | 0.40% | 3.2 | Gage |
| 104 | True position of minor diameter to hole diameter | | 0.40% | 3.2 | Gage |
| Minor | | | | | |
| 201 | Height of insulator | | 0.65% | 3.2 | Gage |
| 202 | Evidence of poor workmanship | | 1.00% | 3.6 | Visual |

NOTES:





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## CLASSIFICATION OF DEFECTS & TESTS



| PARAGRAPH | TITLE | | SHEET 1 OF 2 | | DRAWING NUMBER |
|---|---|---|---|---|---|
| 4.4.2.7 | Nose Crush Switch | | | | 9220940 |
| | | | | | NEXT HIGHER ASSEMBLY |
| CATEGORY | EXAMINATION OR TEST | NO. OF SAMPLE UNITS | AQL OR 100% | REQUIREMENT PARAGRAPH | PARAGRAPH REFERENCE /INSPECTION METHOD |
| Critical | None defined | | | | |
| Major | | | | | |
| 101 | Electrical continuity between inner cone subassembly and end of red wire | | 100% | 3.3.1.2 | 4.5.2.1.b |
| 102 | Electrical continuity between front of ogive, missile subassembly and end of black wire | | 100% | 3.3.1.1 | 4.5.2.1.b |
| 103 | Insulation resistance | | 100% | 3.3.2 | 4.5.2.2 |
| 104 | Dimension - outside of ogive to inside of inner cone at apex | | 0.40% | 3.2 | Gage |
| 105 | Any rivet connection, inner cone to ogive, not secured properly | | 0.40% | 3.2 | Visual/Manual |
| 106 | Assembly incomplete - any part missing or defective | | 0.40% | 3.2 | Visual |
| 107 | Length of wires, min. | | 0.40% | 3.2 | Gage |
| 108 | Wire routing improper | | 0.40% | 3.2 | Visual |
| 109 | Paint adhesion - knife test on: test panels | ** | 1.50% | 3.4.2 | Testing sequence - see Fig. 3/4.5.3.2.1 |
| 110 | Paint adhesion - tape test on: test panels | ** | 1.50% | 3.4.2 | Testing sequence - see Fig. 3/4.5.3.2.1 |
| Minor | | | | | |
| 201 | Identification missing or illegible | | 0.65% | 3.2 | Visual |
| 202 | Outer nose tip surface of ogive not free of paint | | 0.65% | 3.2 | Visual |
| 203 | Inner periphery at warhead body interface not free of paint | | 0.65% | 3.2 | Visual |

NOTES:

** Sample size per MIL-STD-105, special Inspection Level S-3.





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# CLASSIFICATION OF DEFECTS & TESTS

| PARAGRAPH | TITLE | | SHEET 2 OF 2 | | DRAWING NUMBER |
|---|---|---|---|---|---|
| 4.4.2.7 | Nose Crush Switch | | | | 9220940 |
| | | | | | NEXT HIGHER ASSEMBLY |
| CATEGORY | EXAMINATION OR TEST | NO. OF SAMPLE UNITS | AQL OR 100% | REQUIREMENT PARAGRAPH | PARAGRAPH REFERENCE /INSPECTION METHOD |
| 204 | Protective finish inadequate, scratched or damaged | | 0.65% | 3.6.2 | Visual |
| 205 | Evidence of poor workmanship | | 1.00% | 3.6 | Visual |

NOTES:





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## CLASSIFICATION OF DEFECTS & TESTS



| PARAGRAPH | TITLE | | SHEET 1 OF 1 | | DRAWING NUMBER |
|---|---|---|---|---|---|
| 4.4.2.8 | Wireway | | | | 9216784 |
| | | | | | NEXT HIGHER ASSEMBLY 9233178 |
| CATEGORY | EXAMINATION OR TEST | NO. OF SAMPLE UNITS | AQL OR 100% | REQUIREMENT PARAGRAPH | PARAGRAPH REFERENCE /INSPECTION METHOD |
| Critical | None defined | | | | |
| Major | | | | | |
| 101 | Angles of curvature at both ends and overall form improper | | 0.40% | 3.2 | Gage |
| 102 | Inside of wireway obstructed or deformed to prevent free passage of cable | | 0.40% | 3.2 | Gage |
| Minor | | | | | |
| 201 | Overall length | | 0.65% | 3.2 | Gage |
| 202 | Distance between point of ogive curvature and warhead end | | 0.65% | 3.2 | Gage |
| 203 | Length of bent end fitting into warhead | | 0.65% | 3.2 | Gage |
| 204 | Protective finish scratched, inadequate, or damaged | | 0.65% | 3.6.2 | Visual |
| 205 | Evidence of poor workmanship | | 1.00% | 3.6 | Visual |

NOTES:

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete



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## CLASSIFICATION OF DEFECTS & TESTS

MIL-S-50218B(AR)

| PARAGRAPH | TITLE | SHEET 1 OF 1 | DRAWING NUMBER |
|---|---|---|---|
| 4.4.2.9 | Other Components | | See BELOW |
| | | | NEXT HIGHER ASSEMBLY |

| CATEGORY | EXAMINATION OR TEST | NO. OF SAMPLE UNITS | AQL OR 100% | REQUIREMENT PARAGRAPH | PARAGRAPH REFERENCE /INSPECTION METHOD |
|---|---|---|---|---|---|
| | Dwg. No. Document Nomenclature<br>9233180 Washer, Nonmetallic<br>9233181 Washer, Flat<br>MS35333-37 Washer, Lock, Flat<br>MS25036-101 Terminal Lug<br>9317983 Grommet:Rubber<br>MS16535-155 Rivet - Tubular, Oval Head<br>MS16535-163 Rivet - Tubular, Oval Head<br>9326893 Washer, Flat<br>The contractor or vendor shall supply a certificate that the materials used conform to the drawing requirements. | | | | |
| Critical | None defined | | | | |
| Major<br>101 | Dimensions improper | | 0.40% | 3.2 | Gages |
| Minor<br>201<br>202 | Protective finish inadequate (if applicable)<br>Evidence of poor workmanship | | 0.65%<br>1.00% | 3.6.2<br>3.6 | Visual<br>Visual |

NOTES:



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4.4.3 Testing

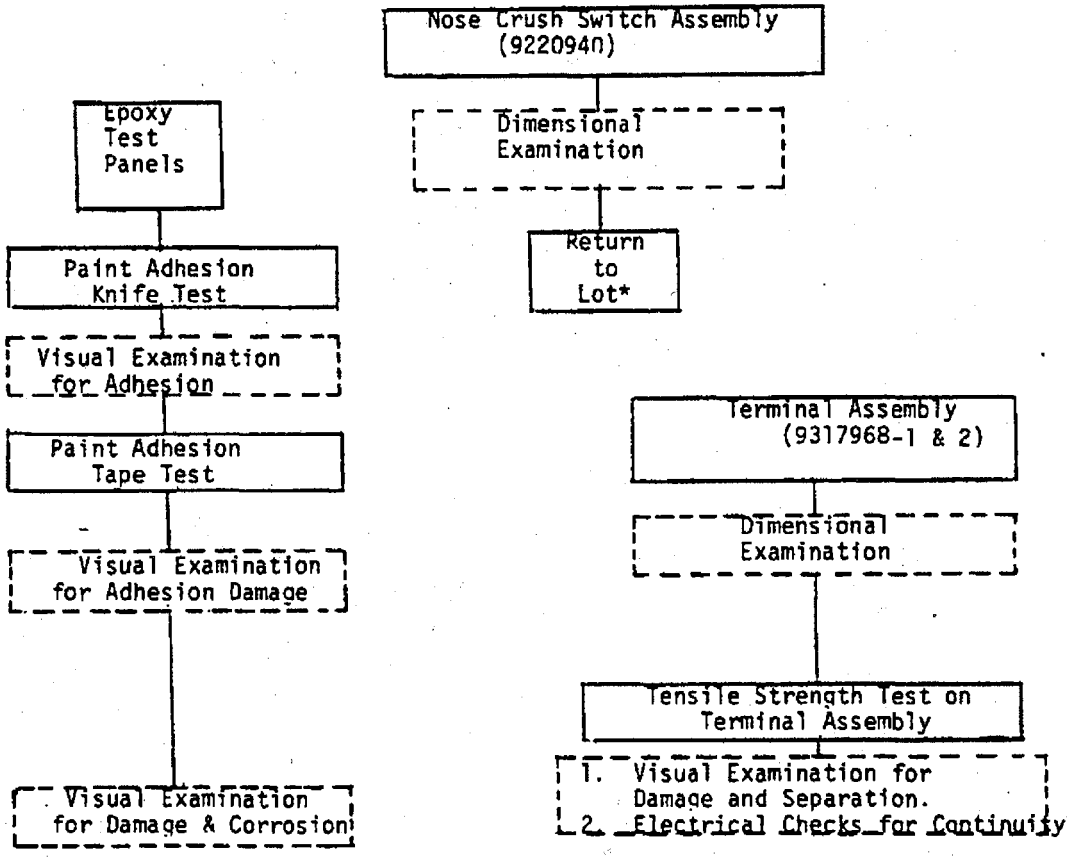
4.4.3.1 Testing.-Testing is described in the First Article and Quality Conformance tables.

4.4.4 Inspection equipment.-The inspection equipment required to perform the examinations and tests prescribed herein is described in the "Paragraph Reference/Inspection Method" column in the tables starting with paragraph 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and 6.3 herein.



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PRODUCTION LOT ACCEPTANCE TEST




NOTE: *None of the tactical units indicated above will be subjected to the destructive testing for corrosion resistance of the projectile finish, and if not damaged during examination or found defective, may be returned to the lot.

FIGURE 3



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4.5 Methods of inspection

4.5.1 Materials, components and processes.-Compliance with all requirements of section 3 of this specification shall be ascertained by current and continuing examination of inspection and test data to determine that all components (parts, subassemblies and materials) have been inspected and tested and found to comply with their respective drawing and specification requirements, and that all specified manufacturing processes have been followed.

4.5.1.1 Visual, mechanical and electrical inspections.-The visual manual, mechanical and electrical inspections and tests shall verify compliance with the requirements of sections 3 and 5 of this specification in accordance with 4.4.2 herein and Table I.

4.5.2 Electrical continuity and insulation resistance of nose crush switch components and assemblies.

4.5.2.1 Electrical continuity.-Continuity measurements shall be made of each of the Terminal Assembly (dwg. 9317968-1 & -2), and Nose Crush Switch Assembly (dwg. 9220940), using a milliohmeter.

a. For terminal assemblies - Contact shall be made between the terminal lug and the ends of the wire.

b. For the nose crush switch assembly - Contact shall be made between the front of the ogive missile subassembly and the end of the black wire and between any exposed metal surface of the inner cone subassembly and the end of the red wire.

Any assembly on which the resistance measurements of either of these circuits exceeds the maximum resistance requirement shall be classed defective and removed from the lot.



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4.5.2.2 Insulation (open circuit) resistance of the nose crush switch assembly (dwg. 9220940).-Insulation resistance measurements shall be taken between the ends of the red and black wire. Resistance shall be measured under in impressed voltage of 500 volts plus or minus (±) 50 volts. D.C. for two (2) seconds, min. Any assembly exhibiting a lower insulation resistance measurement than the requirement shall be classed defective and removed from the lot.

4.5.3 Protective finishes

4.5.3.1 Corrosion resistance.-The protective finish surfaces of the test panels representing the missile ogive and the inner cone shall be subjected to a 5 percent salt spray test in accordance with MIL-C-5541 and the applicable Method 6061 of Federal Test Method Standard No. 141, for a period of 100 hours, using test equipment as specified therein or equivalent approved equipment. Any test panel on which the chemical film protective finish fails to meet the requirements shall be classed defective and the corresponding parts represented by the failing test panel(s) shall be rejected.

4.5.3.2 Paint adhesion.

4.5.3.2.1 Knife and tape tests.-The knife test and tape test for paint adhesion properties of the protectively finished surface to be tested shall be performed in accordance with the procedure in MIL-C-5541 and the applicable Methods 6301.1 and 6304.1 of Federal Test Method Standard No. 141, using test equipment and materials as specified therein or their approved equivalents. Visual examination for paint adhesion failure shall be made after each test. If the paint film forming the protective finish on any of the assemblies or specimen panels tested fails to meet the requirements, the assembly or specimen panel shall be classed defective.

"4.5.4 Tensile strength of terminal assembly.-The terminal assembly shall be subjected to a tensile strength test in accordance with MIL-E-45782, para. 3.4.2.1 and 4.6.2.1.2 except the speed of the holding head shall not exceed 10 inches per minute. (Destructive test)."



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4.5.5 Stamped components.-At the start of production from a new set of dies, or after repair or adjustment to such dies, consecutively produced units from each set of dies shall be physically and dimensionally examined until the dies produce units of product that conform to all applicable drawing requirements. A first article shall then be submitted consisting of all units produced in the prescribed number of consecutive stamping operations from each set of dies (see 4.3). During production, the stamping procedures shall be monitored for uniformity of the stamping operation and quality conformance of product. When a Classification of Defects under 4.4.2 is specified for a stamped component, lot acceptance shall be in accordance thereto, and will serve as a further check on the stamping process. Suspected deviations in any die or the product shall be investigated to determine if corrective action, die repair, or adjustment is required. When any corrective action is undertaken which may have an effect on the dimensional or physical characteristics of the product, production shall not be resumed until consecutively produced units from such dies produce units of product that conform to all applicable drawing requirements.

4.5.6 Molded components.-At the start of production from a new mold, or repair to a mold or cavity, consecutively produced units from each mold or cavity shall be physically and dimensionally examined until four (4) consecutive units meet all applicable drawing requirements. After adjustment or repair, those dimensions affected by the adjustment or repair shall be examined. During production, the molding procedures shall be monitored for uniformity of molding practices and quality conformance of product. When a Classification of Defects under 4.4.2 is specified for a molded component, lot acceptance shall be in accordance thereto, and will also serve as a continuing check on the molding process. Suspected deviations in the mold or product shall be examined to determine if corrective action or mold repair is required.

## 5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging, Level C.-The parts shall be packaged and packed in accordance with the requirements of MIL-STD-1169.



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5.2 Marking.-Marking shall be in accordance with the requirements of MIL-STD-1169.

6. NOTES

6.1 Intended use.-The nose crush switch assembly covered by this specification is a major component of the Dragon Warhead Section, Guided Missile, HE, M225.

6.2 Ordering data.-See MIL-A-48078.

6.3 Submission of inspection equipment for design approvals.-See MIL-A-48078. Submit designs as required to Commander, US Army Armament Research and Development Command, ATTN: DRDAR-QAR-I, Dover, NJ 07801.

6.4 Drawings.-Drawings listed in Section 2 of this specification under the heading US Army Armament Research and Development Command (ARRADCOM) may also include drawings prepared by, and identified as, Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data orginally prepared by these activities is now under the cognizance of ARRADCOM.

6.5 Changes from previous issue.-Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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